

The Calibration Paradox in Fellegi-Sunter Processes

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Abstract

Probabilistic record linkage is built on the Fellegi–Sunter model, in which a decision depends on three coupled quantities: the per-field match and non-match probabilities m, u , the prior match probability λ , and a decision threshold τ . In a two-stage entity-resolution pipeline, re-estimating these quantities from data — especially expectation–maximisation (EM) with a freely fitted prior — repeatedly produced *worse* results at a fixed τ than leaving the parameters at their untrained defaults, while supervised calibration could either help or hurt depending on the data. We call this the calibration paradox: improving the fidelity of the evidence model can degrade the measured operational outcome. We show that the mechanism is not a flaw in training but a coordination failure. Writing the decision rule in log-odds separates the evidence weights $w = \log_2(m/u)$ from a prior bias $b = \log_2(\lambda/(1 - \lambda))$; the effective match-weight threshold is $w_\tau^* = \log_2(\tau/(1 - \tau)) - b$. Retraining changes w , while a freely fitted prior changes b , so a fixed τ selects a different operating point in every configuration. We derive the paradox under this decomposition, distinguish prior coupling from a genuine m/u residual, and give a tandem (λ, τ) calibration procedure, together with an analysis of how threshold choice is sensitive to prior misspecification. Empirical evidence from a 100,000-record duplicate-bearing benchmark, a joint (λ, τ) surface, a real-schema NC-voter replication, and a smoothing regularisation study illustrates both the failure and its repair.

1 Introduction

Entity resolution seeks to determine whether two records refer to the same real-world entity. The dominant computational recipe first retrieves a small set of candidate records (blocking) and then decides for each candidate pair by probabilistic linkage [1, 4]. The probabilistic stage computes a match probability from the evidence of agreement across fields and classifies a pair as a match when that probability reaches a threshold.

In building and validating a two-stage pipeline [7], we observed an unexpectedly robust phenomenon we call the *calibration paradox*: re-estimating the model’s match and non-match probabilities m and u from the data — which should *improve* calibration — can *worsen* the reported outcome at a fixed decision threshold τ . The effect is sharpest and most robust for unsupervised learning (EM) with a freely re-estimated prior λ : on some configurations EM degraded an F1 of roughly 0.98 to below 0.9. On a synthetic duplicate-bearing benchmark, supervised calibration also trailed the untrained defaults at a fixed τ ; on harder real-schema data, however, supervised calibration can *help* (Section 5.3). The paradox is therefore not “always keep the defaults,” but that calibration and the decision rule are coupled and must be reconciled together.

This seems paradoxical: training the parameters on data is the standard remedy for poor record-linkage performance, and the literature reports that appropriately estimated m/u and priors improve linkage [2, 3]. The resolution, we argue, is that the decision problem is not governed by m/u alone. It is governed by a *bundle* of three coupled quantities — the evidence weights derived from m/u , the match prior λ , and the decision threshold τ — and these are not interchangeable.

A practitioner who retrains m/u and leaves τ fixed, or who lets the same fitting procedure also estimate λ freely, has not improved the decision; it has chosen a different operating point, usually a worse one.

The contribution of this paper is a clean theoretical account of the paradox and a practical repair. We formalise the Fellegi–Sunter decision rule in log-odds (Section 2), which decomposes the posterior into an evidence term (from m/u) and a bias term (from λ), with τ selecting a weight threshold. We then derive the paradox as a coordination failure of this decomposition and separate its two mechanisms: *prior* coupling and a *genuine residual* from over-weighted partial-similarity levels (Section 3). We give a tandem (λ, τ) calibration procedure (Section 4) and demonstrate it empirically on a large duplicate-bearing benchmark and a joint (λ, τ) surface (Section 5). Section 6 relates the findings to the literature and Section 7 concludes.

2 Background: the Fellegi–Sunter model and its three parameters

Consider a candidate pair of records and a comparison vector γ describing whether each field agrees (exactly or to a graded degree). The Fellegi–Sunter model [1] posits two latent classes: the pair is either a true match M or a non-match U . Let

$$m(\gamma) = P(\gamma \mid M), \quad u(\gamma) = P(\gamma \mid U). \quad (1)$$

Under conditional independence across fields, $m(\gamma)$ and $u(\gamma)$ factor into per-field, per-level probabilities, so the log-likelihood-ratio (in bits) is additive:

$$w(\gamma) = \log_2 \frac{m(\gamma)}{u(\gamma)} = \sum_k w_k(\gamma_k), \quad w_k = \log_2 \frac{m_k}{u_k}. \quad (2)$$

The prior probability that an arbitrary pair matches is

$$\lambda = P(M). \quad (3)$$

By Bayes’ theorem the posterior probability that the pair is a match is

$$P(M \mid \gamma) = \frac{\lambda m(\gamma)}{\lambda m(\gamma) + (1 - \lambda) u(\gamma)}. \quad (4)$$

A practitioner declares the pair a match when the posterior reaches a threshold:

$$\text{decision}(\gamma) = \begin{cases} \text{match} & P(M \mid \gamma) \geq \tau, \\ \text{non-match} & \text{otherwise.} \end{cases} \quad (5)$$

The three quantities in (5) and (4) play different roles:

m, u describe the *evidence*: how discriminating agreement or disagreement on each field is. They enter only through the weight w .

λ describes the *prior base rate*: how many of the pairs examined are expected to be true matches. It is a global bias applied to every pair.

τ describes the *target*: the posterior confidence at which we are willing to declare a match; equivalently it encodes the relative cost of a false positive versus a false negative [2].

2.1 Log-odds decomposition

It is convenient to work in log-odds (base 2). For any probability p , write $\text{logit}_2(p) = \log_2 \frac{p}{1-p}$. Taking log-odds of (4),

$$\text{logit}_2(P(M \mid \gamma)) = w(\gamma) + \underbrace{\log_2 \frac{\lambda}{1-\lambda}}_b, \quad (6)$$

where we have defined the *prior bias*

$$b = \log_2 \frac{\lambda}{1-\lambda}. \quad (7)$$

Thus the posterior log-odds is the evidence weight $w(\gamma)$ plus a constant prior bias b that is identical for every pair.

Writing the decision threshold in the same units, $t = \text{logit}_2(\tau)$, the decision rule (5) becomes

$$\text{match iff } w(\gamma) \geq t - b = \log_2 \frac{\tau}{1-\tau} - \log_2 \frac{\lambda}{1-\lambda} =: w_\tau^*(\lambda). \quad (8)$$

Equation (8) is the key object of this paper. It shows that a fixed posterior threshold τ does *not* correspond to a fixed evidence threshold: the effective match-weight threshold w_τ^* is a function of the prior λ . Raising λ increases b , lowers w_τ^* , and therefore admits weight values that were previously below threshold; lowering λ has the opposite effect.

3 The calibration paradox: theoretical derivation

3.1 Statement

Let $\theta = (m, u, \lambda, \tau)$ be a configuration of the model. For a set of candidate pairs, denote the confusion matrix under θ and its consequent F1(θ), or more generally any loss $\mathcal{L}(\theta)$. The calibration paradox is the observation that a configuration θ' whose *evidence parameters* (m', u') are better fitted to the data than (m, u) (in the sense of higher likelihood, or trained on labelled duplicates) frequently satisfies

$$\mathcal{L}(\theta') > \mathcal{L}(\theta), \quad (9)$$

i.e. the “better-calibrated” evidence *degrades* the decision outcome when the threshold is kept fixed. In our pipeline the paradox is sharpest when training also refits the prior, so that λ moves as well.

3.2 Mechanism 1: prior coupling

Suppose retraining leaves the evidence weights unchanged but shifts the prior from λ to λ' . By (6), every pair’s posterior log-odds changes by

$$\Delta b = \log_2 \frac{\lambda'/(1-\lambda')}{\lambda/(1-\lambda)}, \quad (10)$$

and, by (8), the effective weight threshold moves by $\Delta w_\tau^* = -\Delta b$. Concretely, starting from the default $\lambda = 10^{-4}$ ($b \approx -13.3$ hits) an EM fit that raises λ to 7.2×10^{-3} ($b \approx -7.1$ bits) reduces the effective threshold by ≈ 6.2 bits. Every candidate pair whose weight lies within 6.2 bits below the old boundary now crosses it. Non-match pairs are typically far more numerous and concentrated near the boundary (they include partial agreement), so the effect is a disproportionate inflow of false positives: specificity collapses while recall is barely affected.

The prior is therefore not *merely* a base rate; it is a global shift of the whole posterior distribution. Because the default λ and the default τ were chosen to be mutually consistent, leaving τ fixed while λ drifts selects a different, usually worse, region of the precision–recall curve. We label this *prior coupling*: the prior and the threshold must move together.

3.3 Mechanism 2: the evidence residual

Even with the prior frozen ($b' = b$), retraining m/u can perturb the distribution of weights. Supervised calibration on labelled duplicates, or EM on a population that contains such duplicates, tends to assign higher m to the *partial-similarity* levels (e.g. an exact first-name match with a small edit on the last name, or a near-matching address). These partial levels are exactly the ones that many *non-match* pairs also occupy, so a higher m there raises w_k for those levels. Non-match pairs therefore move *right* in the weight distribution, increasing the density of non-matches near the decision boundary. For a fixed threshold τ this lowers precision and specificity (extra false positives), even though the evidence model is arguably “more accurate.”

We call this the *genuine residual*: it survives prior anchoring, and in our data it keeps even the jointly re-optimised trained model slightly below the untrained defaults. It is a substantive statement about which levels should be rewarded, not a numerical artefact.

3.4 The coalescence: fixed τ is not a fixed operating point

Combining both mechanisms and expanding the log-odds for a pair p ,

$$\Delta \logit_2(P(M \mid \gamma_p)) = (w'(\gamma_p) - w(\gamma_p)) + (b' - b). \quad (11)$$

A pair crosses the threshold in configuration θ' but not in θ whenever $\Delta \logit_2(P(M \mid \gamma_p)) \geq \logit_2(\tau) - \logit_2(P(M \mid \gamma_p))$. Because the evidence shift $w' - w$ is concentrated where the non-match density is largest, and the bias shift $b' - b$ moves the whole mass, the set of newly-admitted pairs is overwhelmingly false positives. The “calibration paradox” is thus a *coordination* failure: (m, u, λ, τ) are not four free dials but one coupled decision rule, and turning one dial without the others moves the operating point to an inferior location.

4 Joint (λ, τ) calibration

The theory of Section 3 immediately suggests the repair: never change the evidence m/u without re-anchoring the prior λ and re-selecting the threshold τ together, and validate the combination on held-out data. We formalise this as a tandem procedure.

- Step 0.** Baseline prior $\lambda_0 = 10^{-4}$ (a sensible default given a sparse duplicate base rate).
- Step 1.** Estimate a *principled* prior rather than a free one. With labels, $\lambda = \frac{\text{labelled match rate}}{\text{blocking recall}}$, the base rate in random-pair space. Without labels, use a blocking-adjusted estimator [3] that divides an assumed expected-match count by the blocking coverage; if no defensible rate exists, keep λ_0 .
- Step 2.** Fit m/u with the prior *fixed* (e.g. EM with λ held at the Step 1 value, or supervised calibration), so that training affects only the evidence weights.
- Step 3.** On a validation set, jointly select $(\lambda, \tau) \in \{\lambda_0, \lambda_{\text{est}}\} \times 10^{-k} \times [0.5, 0.999]$ optimising the objective: F1, or a decision cost $C_{FP}FP + C_{FN}FN$, or the expected-F1 derived from calibrated posteriors.

Step 4. Coherence guard: reject solutions whose mean posterior probability of *non-match* pairs is large (in our data, EM’s reached ≈ 0.55). This diagnostic separates prior coupling (fixable by lowering λ) from the genuine m/u residual (requires changing the comparison model).

Step 5. Freeze (λ^*, τ^*) on validation and evaluate once on a held-out test set.

5 Empirical demonstration

5.1 Setup

We used the two-stage pipeline from the companion entity-resolution paper [7]: records are embedded with a sentence-transformer model, blocked by cosine nearest-neighbour search, and linked with a Fellegi–Sunter scorer (Splink) using Jaro–Winkler comparisons for names and address, a date-of-birth comparison, and an email comparison. All experiments were run across a 100,000-record base population with a 3% duplicate-bearing augmentation, producing 6,000 labelled queries (positives are near-duplicate variants, negatives are held-out records).

Three parameter configurations are compared: (i) untrained defaults, with $\lambda = 10^{-4}$; (ii) supervised calibration of m/u from the labelled duplicate pairs, $\lambda = 10^{-4}$; (iii) EM with a freely estimated prior (which produced $\lambda \approx 7.2 \times 10^{-3}$), and an ablation of (iii) with λ reset to 10^{-4} . Table 1 reports the outcome at the fixed decision threshold $\tau = 0.85$.

Table 1: Duplicate-bearing benchmark (100,000 base records, 6,000 queries, $\tau = 0.85$). The trained configurations degrade F1 at the fixed threshold; the EM model degraded most, and its prior only explains part of the loss.

Configuration	Precision	Recall	Specificity	F1	λ
Untrained defaults	90.1%	98.1%	89.2%	93.9%	10^{-4}
Supervised (m/u , λ fixed)	75.1%	98.5%	67.3%	85.2%	10^{-4}
EM (free prior)	56.9%	98.3%	25.4%	72.1%	7.2×10^{-3}

The free-prior EM configuration is the clearest quantitative instance of the paradox: the same model that recovered a fraction of the unweighted prior’s recall errors collapsed precision and specificity. In an ablation on the same synthetic duplicate-bearing population we reset the EM prior to 10^{-4} and re-tuned τ ; EM’s F1 recovered from 0.826 to 0.968, and the mean posterior probability of non-match pairs fell from 0.549 to 0.328. This is consistent with Mechanism 1 (prior coupling) being the dominant term: holding λ at the default removes most of the apparent failure.

5.2 Joint (λ, τ) surface

To show that the paradox is an operating-point effect and not a property of any single threshold, we swept the prior and the threshold jointly for the untrained and EM models (Table 2). The untrained defaults retain F1 above 0.97 across the whole region; the EM model only reaches comparable F1 near low λ and high τ , and even its joint optimum (0.955) stays below the untrained optimum (0.997). This is exactly the prediction of Section 3: prior coupling is removed by sweeping, but the genuine m/u residual remains.

Table 2: Joint F1 over (λ, τ) for untrained and EM configurations on the duplicate-bearing population. Entries are F1 (percent).

λ	Untrained		EM	
	$\tau = 0.85$	$\tau = 0.95$	$\tau = 0.85$	$\tau = 0.95$
10^{-4}	98.4	99.7	93.3	94.2
10^{-3}	97.1	97.1	81.7	84.7
10^{-2}	95.8	96.4	81.0	81.4

5.3 Real-data validation on NC-voter

To test whether the paradox is an artefact of the synthetic generator, we replicated the comparison on the North Carolina voter-registration file: a public, externally pre-deduplicated dataset with no full birth date or email. Because authentic duplicate pairs are unavailable, the evaluation mirrors the pipeline paper’s protocol: positives are mutated duplicates of records held in the index and negatives are mutated held-out records, scored at a fixed $\tau = 0.85$ (Table 3).

Table 3: Mutated NC-voter resolution at fixed $\tau = 0.85$ (3,000 in-index, 1,500 positive, 1,500 negative queries). Date-of-birth and email comparisons are unavailable.

Configuration	Precision	Recall	Specificity	F1
Untrained defaults	100.0%	85.6%	100.0%	92.2%
Supervised m/u	99.9%	89.3%	99.9%	94.3%
EM (free prior)	73.7%	52.9%	81.1%	61.6%
EM, prior reset to 10^{-4}	100.0%	24.3%	100.0%	39.1%

Two findings qualify the synthetic result. First, on this harder real schema supervised calibration does *help*: it raises recall (85.6% to 89.3%) at virtually no cost to precision (100.0% to 99.9%). The “untrained defaults win” claim in the synthetic benchmark is therefore not universal; where partial agreement is genuinely informative, trained m/u can be better. Second, the *unsupervised* (EM) result remains catastrophic — F1 collapses from 92.2% to 61.6% — and, notably, resetting the EM prior to 10^{-4} makes it *worse* (39.1%) by crushing recall. On this data EM’s fitted m/u are themselves unreliable, so prior anchoring alone cannot repair them. The robust, general conclusion is that freely fitting EM parameters is dangerous without joint calibration, whereas supervised calibration transfers to a real schema and can be beneficial; the two are not interchangeable remedies.

5.4 Smoothing and regularisation

A natural question is whether the genuine residual stems from noisy per-level probability estimates, which regularisation could fix. If level k of a comparison has n_k observed occurrences among a total of N training pairs over K levels, the empirical m -probability is $\hat{m}_k = n_k/N$. Laplace (additive) smoothing, equivalently a Dirichlet prior with concentration α , replaces it by

$$\hat{m}_k^{(\alpha)} = \frac{n_k + \alpha}{N + K\alpha}, \quad (12)$$

which pulls the estimate toward the uniform value $1/K$ and removes zero counts. Because the residual mechanism of Section 3 is that trained models over-weight partial-similarity levels, and those levels are precisely the ones where n_k is large, one might expect larger α to reduce the residual.

Table 4 tests this on the supervised calibration of the 50,000-reference benchmark at fixed τ . Increasing α from 0.5 to 50 barely moves the supervised F1 (94.4% to 93.6%), leaving a persistent gap of roughly four points below the untrained defaults ($\approx 98\%$). The residual is therefore not an artefact of level-count variance or zero counts; it reflects a structural modelling choice — the trained model rewards partial-similarity agreement more than the default weight scheme does — and smoothing the level probabilities toward uniform does not reverse that choice. We therefore recommend Laplace/Dirichlet smoothing as a variance remedy (and use it in our supervised estimator by default), but stress that it does not, by itself, close the paradox; the repair remains the joint (λ, τ) calibration of Section 4.

Table 4: Supervised calibration of the synthetic benchmark at fixed $\tau = 0.85$ under Laplace smoothing α . The untrained defaults give F1 $\approx 98\%$ throughout.

Smoothing α	0.5	5	50
Supervised F1	94.4%	93.8%	93.6%

5.5 Sensitivity of the threshold to prior misspecification

Equation (8) makes the coupling explicit: $w_\tau^* = t - b$, with $t = \log_2 \frac{\tau}{1-\tau}$ and $b = \log_2 \frac{\lambda}{1-\lambda}$. If the true prior is λ but the pipeline uses a misspecified $\tilde{\lambda} = c\lambda$, the bias changes by

$$\Delta b = \log_2 \frac{\tilde{\lambda}/(1-\tilde{\lambda})}{\lambda/(1-\lambda)} \approx \log_2 c \quad (\lambda \ll 1). \quad (13)$$

To preserve the same effective threshold, τ must be adjusted by the same number of bits: a factor c in the prior requires an offset of $\approx \log_2 c$ in $\text{logit}_2(\tau)$. In fully unsupervised settings λ cannot be estimated reliably, so this relation bounds how much the operating point can move for a given prior error.

Table 5 quantifies the consequence: the best F1 attainable after re-optimising τ at each prior. The untrained model is stable (best F1 0.98–1.00) across two orders of magnitude in λ ; the EM model is highly sensitive, its attainable F1 falling from 0.96 at $\lambda = 10^{-5}$ to 0.81 at $\lambda = 10^{-2}$. Our guidance for the unsupervised setting is therefore: (i) prefer the default or blocking-adjusted prior rather than a free EM estimate; (ii) select τ on a validation split over a grid of candidate λ ; and (iii) report the range of F1 across that grid as an explicit sensitivity bound. A wide range (as for EM) is a warning that the configuration is unsafe without labelled data.

Table 5: Best F1 attainable after re-optimising τ at each assumed prior, on the duplicate-bearing population. The optimal τ for every EM row is 0.98.

Assumed λ	10^{-5}	10^{-3}	10^{-2}
Untrained, best F1	0.99	0.99	0.98
EM, best F1	0.96	0.91	0.81

6 Related work

The Fellegi–Sunter model [1] already framed linkage as a decision problem with error-rate targets, and Winkler’s line of work formalised optimal decision rules and frequency-based parameter esti-

mation [2, 3]. Christen’s monograph [4] surveys pragmatic threshold and calibration practice. The precision–recall and threshold-selection literature [5] treats the operating point as a tunable trade-off but, as far as we know, does not make explicit that m/u , λ , and τ form a single coupled decision rule whose components cannot be changed independently. The contribution here is that explicit log-odds decomposition ((6)–(8)), the identification of prior coupling as the dominant mechanism, and the tandem calibration procedure.

7 Discussion and conclusion

The calibration paradox is the statement that training the evidence parameters m/u can worsen a decision that already includes a prior λ and a threshold τ . The log-odds decomposition shows why: the posterior log-odds is the sum of the evidence weight w and a prior bias $b = \log_2 \frac{\lambda}{1-\lambda}$, and the decision compares this to $t = \log_2 \frac{\tau}{1-\tau}$. Retraining m/u moves the evidence distribution; a freely estimated prior moves the bias; a fixed τ holds the target fixed. Changing any one without the other two relocates the operating point.

Two distinct mechanisms drive the observed drop. *Prior coupling* is a global bias shift: an EM prior inflated from 10^{-4} to 7.2×10^{-3} lowers the effective weight threshold by about 6 bits and admits a flood of false positives; re-anchoring the prior recovers most of the loss on the synthetic benchmark. *The genuine residual* is a shift of the non-match weight distribution caused by over-weighting partial-similarity levels, which survives prior anchoring and keeps jointly re-optimised trained models slightly below the defaults on the synthetic data. Real-data replication (Section 5.3) refines this: supervised calibration can improve outcomes on harder schemas (it lifted NC-voter F1 from 92.2% to 94.3% at fixed τ), whereas freely fitted EM remains catastrophically miscalibrated. Smoothing the level probabilities (Section 5.4) is a valuable variance remedy but does not, by itself, close the residual; and the threshold is explicitly sensitive to prior misspecification (Section 5.5), so a wide F1 range across candidate priors is a warning sign in unsupervised settings.

The practical message is methodological: in a Fellegi–Sunter process, the triple $(m/u, \lambda, \tau)$ is a *coherent bundle*. Calibration is only meaningful when the evidence, the prior bias, and the decision threshold are estimated and selected together, and then evaluated on held-out data. Retraining m/u while leaving λ and τ unchanged is not a safe optimisation; it is a hidden change of operating point, and—for unsupervised calibration especially—usually a worse one.

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Use of Artificial Intelligence

During the preparation of this work the author used an AI coding and writing assistant (the DeepSeek V4 Flash 0731 large language model) to assist with drafting and editing the manuscript and preparing the experiments. The tool was not listed as an author. The theoretical derivation and all reported results were reviewed and verified by the author, who takes full responsibility for the content, including any remaining errors and/or omissions.